

# Factory Reallocation and Shipping Optimization System using Predictive Modeling and Decision Intelligence

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**Abstract**— This research proposes a predictive decision-support system for optimizing factory allocation and shipping performance at Nassau Candy Distributor. Traditional static factory assignments often increase delivery time and reduce operational flexibility. Using transactional sales and logistics data, machine learning models were developed to predict shipping lead time and evaluate alternative factory assignments. Scenario simulation and weighted decision scoring were incorporated to recommend optimal factory allocations while balancing operational efficiency and profitability. Experimental results demonstrate that the Random Forest model provides reliable predictions and that selected reassignment scenarios can reduce lead times without significantly affecting profit. The proposed framework illustrates how predictive analytics and decision intelligence can improve logistics planning and support data-driven supply chain management.

**Keywords**— Predictive Modeling, Factory Optimization, Machine Learning, Decision Intelligence, Supply Chain

## I. INTRODUCTION

Factory allocation plays a significant role in determining shipping efficiency, operational cost, and customer satisfaction. Nassau Candy Distributor traditionally assigns products to factories using static business rules, limiting its ability to respond to changing demand and logistics conditions. This study introduces a predictive decision intelligence framework capable of estimating shipping performance and recommending improved factory allocations. By combining machine learning with scenario simulation, the proposed system enables managers to evaluate multiple allocation strategies before implementing operational changes.

## II. DATASET AND METHODOLOGY

The analysis was conducted using transactional records containing product names, factory assignments, customer regions, shipping modes, sales, units, costs, gross profit, and order and shipment dates. Lead time and profit margin were derived as analytical features. After cleaning the dataset, categorical variables were encoded and numerical variables normalized to prepare the data for predictive modeling. Linear Regression, Gradient Boosting, and Random Forest algorithms were evaluated using RMSE, MAE, and  $R^2$  metrics, with Random Forest selected as the final model because it achieved the best balance between predictive accuracy and interpretability.

## III. RESULTS AND DISCUSSION

The developed simulation engine evaluated multiple factory allocation scenarios for each product by predicting lead time and estimating the financial impact of reassignment. Results showed that several existing factory assignments were already

optimal, while selected product-region combinations achieved measurable lead-time improvements ranging from approximately 2% to 20%. An important strength of the model was its ability to recommend retaining current assignments when reassignment offered no meaningful benefit, thereby avoiding unnecessary operational disruption. The weighted optimization strategy successfully balanced delivery speed and profitability, producing recommendations that were both practical and financially sustainable.

## IV. RECOMMENDATIONS

The organization should integrate predictive factory allocation into routine logistics planning and continuously retrain the model using updated shipment data. Operational teams should evaluate recommendations through controlled pilot implementations before large-scale deployment. Monitoring prediction accuracy and business outcomes over time will further improve confidence in the decision-support framework and enable continuous optimization.

## V. CONCLUSION

This research demonstrates how predictive modeling and scenario-based decision intelligence can transform conventional logistics planning into a proactive optimization process. By combining machine learning with simulation techniques, the proposed system provides actionable recommendations that improve shipping efficiency while maintaining profitability. The framework offers a scalable approach for supporting strategic supply chain decisions and enhancing long-term operational performance.

## REFERENCES

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